

# **RAGme.io: Personal RAG Agent for Web Content**

## **A Comprehensive Overview**

**Maximilien.ai**

# What is RAGme.io?

**RAGme.io** is a personalized agent that uses [Retrieval-Augmented Generation \(RAG\)](#) to process websites and documents you care about, enabling intelligent querying through an LLM agent.

## Core Concept

- **RAG:** Combines document retrieval with AI generation
- **Personal:** Focuses on your specific content and interests
- **Agentic:** Uses LLM agents for intelligent interaction
- **Multi-modal:** Supports web pages, PDFs, DOCX documents, and images
- **Vector Database Agnostic:** Supports multiple vector databases (Milvus, Weaviate, etc.)
- **Production Ready:** Kubernetes deployment with OAuth authentication

# Modern Interface

## Three-Pane Layout with Real-time Features

 **RAGme.ai Assistant**

Vector DB: **weaviate** | Collection: **RagMeDocs**

+ Add Content

Chat History <

#1753977645443  
who is max  
7/31/2025 09:01 AM



Welcome to  **RAGme.ai Assistant!**

I can help you with:

- **Adding URLs** - Tell me URLs to crawl and add to your knowledge base
- **Adding documents (Text, PDF, DOCX, etc.)** - Use the "Add Content" button to add files and structured data
- **Answering questions** - Ask me anything about your documents
- **Document management** - View and explore your documents in the right panel

Try asking me to add some URLs or ask questions about your existing documents!

<



Based on the stored documents, "Max" refers to Maximilien, who appears to be a photographer and blogger. His website features various collections of photographs from his travels, street photography, and portrait shoots. He has shared experiences, such as visiting his homeland of Haiti for his brother's wedding and attending a concert by the band RAM. For more details, you can visit his website at [maximilien.org](https://maximilien.org).

 who is max

# Advanced Document Management

## Interactive Document Dashboard with AI Features

  **RAGme.ai Assistant**

Vector DB: **weaviate** | Collection: **RagMeDocs** + Add Content

**Chat History** <

#1753977645443  
who is max  
7/31/2025 09:06 AM

portrait shoots. He has shared ex-  
periences, such as visiting his  
hometown of Haiti for his brother's  
wedding and attending a concert by  
the band RAM.  
For more details, you can visit his  
website at [maximilien.org](https://maximilien.org).

 what is ragme



RagMe is a document management  
system that allows users to store,  
retrieve, and manage documents ef-  
ficiently. It typically includes fea-  
tures for uploading documents,  
querying stored information, and or-  
ganizing content for easy access.  
Users can interact with the system  
to find specific information within  
the documents, manage collections,

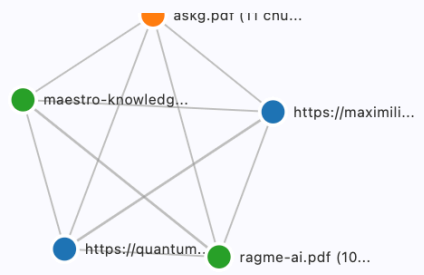
**Documents** Current >

**Documents Overview** Network Graph 

• webpage

• pdf

• PDF



askg.pdf (11 cnu...)

maestro-knowledg...

https://maximili...

https://quantum...

ragme-ai.pdf (10...

askg.pdf 11 chunks

 2025-07-30T22:08:30.269Z |   
Default |  Chunked  
document  
No content available

# Data Visualization & Analytics

## D3.js Powered Interactive Charts

RAGme.ai Assistant

Vector DB: weaviate | Collection: RagMeDocs

+ Add Content

Chat History

#1753977645443  
who is max  
7/31/2025 09:06 AM

ragme-ai.pdf

/CreationDate D:20250727032428Z

/ModDate D:20250727032428Z

Type PDF

Total chunks 10

Is chunked true

Chunk sizes [806,999,1000,738,940,920,999,999,1000,643]

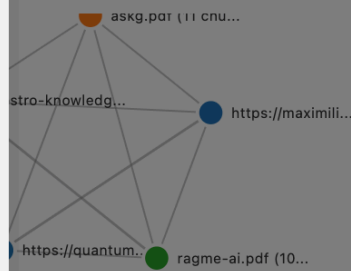
Original filename ragme-ai.pdf

AI Summary

RAGme ai is a personalized agent that leverages Retrieval-Augmented Generation (RAG) to

Current

view Network Graph



```
graph TD; askg.par[askg.par (11 chunks)] --- astro-knowledge[astro-knowledge...]; askg.par --- https_maximili[https://maximili...]; askg.par --- https_quantum[https://quantum...]; astro-knowledge --- https_maximili; astro-knowledge --- https_quantum; https_maximili --- https_quantum; ragme-ai.pdf[ragme-ai.pdf (10...)]
```

11 chunks

269Z |   
unked  
ailable

# Smart Prompt System

## AI-Powered Prompt Suggestions

RAGme.ai Assistant

+ Add Content

Chat History

#1754512464832  
what is ragme  
8/6/2025 09:21 PM

#1754454985343  
what is ragme  
8/6/2025 12:25 PM

#1754427791309  
who is max  
8/5/2025 09:37 PM

watch directory for new files, all with consistent chunking logic.

7. **Chrome Extension:** A one-click capture feature allows users to effortlessly save web pages to their RAGme account for later querying.

**Use Cases:**

- **Intelligent Querying:** Users can ask questions in natural language and receive intelligent, context-driven answers based on their curated content.
- **Document Management:** The system enables effective management of large documents

Documents

Current

Documents Overview

Network Graph

pdf

PDF

webpage

https://rdi.berkeley...

https://maximilien.o...

2506.18511v1.pdf (83...

ragme-ai.pdf (9 chun...

2506.18511v1.pdf

83 chunks

NEW

2025-08-06T04:41:08.364Z

Default

Chunked document

No content available

askg.pdf

11 chunks

2025-08-05T23:57:28.285Z

Default

Chunked document

Recent Prompts & Ideas

who is maximilien?

what is ragme

list my docs

list my recent documents

list my docs

# Latest Features & Capabilities

## 1. Document Processing Pipeline NEW!

- **Batch Processing:** Complete system for processing collections of PDFs, DOCX files, and images
- **Parallel Processing:** Intelligent text chunking with progress tracking
- **PDF Image Extraction:** Comprehensive image analysis with EXIF, AI classification, and OCR
- **Robust Error Handling:** Retry logic with detailed reporting (CSV summaries + .processed files)
- **Command:** `./tools/data_processing.sh /path/to/documents --verbose`

## 2. OAuth Authentication System NEW!

- **Multi-Provider Support:** Google, GitHub, and Apple authentication



# Production-Ready Architecture

## Multi-Service Architecture with Kubernetes Support

```
flowchart TB
    new-ui[New Frontend<br/>Port 8020] --> api[API Server<br/>Port 8021]
    new-ui --> ragme-agent[RagMeAgent<br/>Dispatcher]
    chrome[Chrome Extension] --> api
    subgraph "AI Agent Layer"
        agent-local[File Monitor Local Agent] --> mcp[MCP Server<br/>Port 8022]
        subgraph "Three-Agent System"
            ragme-agent --> functional-agent[FunctionalAgent<br/>Tool Operations]
            ragme-agent --> query-agent[QueryAgent<br/>Content Queries]
            functional-agent --> ragme-tools[RagMeTools<br/>Tool Collection]
            query-agent --> openai[OpenAI LLM]
        end
    end
    mcp --> api
    api --> ragme[RAGme Core]
    ragme --> vector-db[(Vector DB)]
    ragme --> storage-service[Storage Service<br/>MinIO/S3]
    subgraph "Vector Database Layer"
        vector-db --> weaviate-local[(Local Weaviate<br/>Podman)]
        vector-db --> weaviate-cloud[(Weaviate Cloud)]
        vector-db --> milvus[(Milvus Lite)]
    end
    storage-service --> minio[(MinIO<br/>Port 9000)]
    storage-service --> s3[(AWS S3)]
    storage-service --> local-fs[(Local Filesystem)]
    subgraph "Kubernetes Deployment"
        k8s-frontend[Frontend Pods] --> k8s-api[API Pods]
```



# Usage Examples

## 1. Batch Document Processing

```
# Process a collection of documents
./tools/data_processing.sh /path/to/research-papers --verbose

# Query the processed content
"What are the main findings across all research papers?"
```

## 2. Web Content with Authentication

```
# Add authenticated web content
"Crawl my private GitHub repository documentation"

# Query with context
"How do I set up the authentication system?"
```

## 3. Multi-language Content

# API Endpoints

## Content Ingestion

```
# Add URLs with authentication
POST /add-urls
{
  "urls": ["https://private-docs.com", "https://research-paper.org"]
}

# Add JSON content with metadata
POST /add-json
{
  "data": {"content": "..."},
  "metadata": {"source": "...", "language": "en"}
}

# Batch document processing
POST /process-documents
{
  "path": "/path/to/documents",
  "options": {"extract_images": true, "only": true}
```

# Development & Deployment

## Local Development

```
# Quick start with OAuth bypass
./start.sh
# Access: http://localhost:8020

# With full OAuth (requires .env configuration)
# Configure OAuth providers in config.yaml
./start.sh
```

## Kubernetes Deployment

```
# Deploy to GKE
./deployment/scripts/deploy-gke.sh

# Deploy to local Kind cluster
./deployment/scripts/deploy-kind.sh
```



# Data Flow & Processing

## 1. Enhanced Content Ingestion

```
flowchart LR
    A[Web Page/PDF/DOCX/Image] --> B[Authentication Check]
    B --> C[Parser with OCR]
    C --> D[Smart Chunking]
    D --> E[AI Analysis]
    E --> F[Embedding Generation]
    F --> G[Vector DB Storage]
    G --> H[Metadata Caching]
```

## 2. Intelligent Query Processing

```
flowchart LR
    A[User Query] --> B[Language Detection]
    B --> C[Query Classification]
    C --> D[Agent Routing]
    D --> E[Vector Search]
```

# User Interface & Experience

## Modern Three-Pane Layout

- **Left Sidebar:** Chat history with collapsible design
- **Center:** Main chat area with markdown support and real-time updates
- **Right Sidebar:** Document management with D3.js visualizations
- **WebSocket Communication:** Live interactions and real-time updates
- **OAuth Authentication:** Secure login with Google, GitHub, and Apple

## Enhanced Settings Interface

- **Tabbed Organization:** General, Interface, Documents, Chat settings
- **Real-time Configuration:** Changes take effect immediately
- **UI Customization:** Panel widths, visibility, and layout preferences
- **Vector DB Info:** Display database type and collection information

# Key Benefits

## For Individuals

- **Personalized Knowledge Base:** Your own curated content collection with AI insights
- **Multi-language Support:** Process and query content in 80+ languages
- **Intelligent Search:** Natural language queries with context-aware responses
- **Automated Processing:** Batch processing with comprehensive error handling
- **Modern Interface:** Beautiful, responsive web interface with real-time features
- **Secure Access:** OAuth authentication for private content

## For Organizations

- **Enterprise Document Intelligence:** Extract insights from large document collections
- **Research Efficiency:** Rapid analysis with AI-powered summarization
- **Knowledge Discovery:** Find connections across different content sources and



# Technical Highlights

## Performance Optimizations

- **Batch Processing:** Efficient parallel document ingestion with progress tracking
- **AI Summary Caching:** Intelligent caching to prevent redundant AI generation
- **Vector Indexing:** Fast similarity search with configurable thresholds
- **Async Operations:** Non-blocking API responses with WebSocket updates
- **Memory Management:** Optimized storage with base64\_data deduplication
- **Real-time Updates:** WebSocket-based live communication

## Production Features

- **Kubernetes Native:** Full containerization with operator support
- **OAuth Integration:** Secure authentication with multiple providers
- **Health Monitoring:** Comprehensive health checks and auto-recovery

# Future Roadmap

## Phase 1: Infrastructure **COMPLETED**

- [x] ~~Decouple Weaviate dependency~~ **Completed!** Now supports Milvus, Weaviate, and extensible for others
- [x] ~~Add modern frontend UI~~ **Completed!** New three-pane interface with real-time features
- [x] ~~Add local Weaviate support~~ **Completed!** Podman-based local deployment
- [x] ~~Add debugging and monitoring tools~~ **Completed!** Comprehensive log monitoring
- [x] ~~Add OAuth authentication~~ **Completed!** Multi-provider OAuth with JWT sessions
- [x] ~~Add internationalization~~ **Completed!** 80+ language support with automatic detection
- [x] ~~Add batch document processing~~ **Completed!** Full pipeline with progress tracking



## Getting Started Guide

### Quick Setup (Recommended)

```
# Clone and automated setup
gh repo clone maximilien/ragme-io
cd ragme-io
./setup.sh # Automated setup with all dependencies
```

### Manual Setup

```
# Environment setup
uv venv && source .venv/bin/activate
uv sync --extra dev

# Configuration
cp env.example .env
cp config.yaml.example config.yaml
# Edit .env with your API keys
```

# Contributing

For detailed contribution guidelines, see [CONTRIBUTING.md](#).

## How to Help

- **Bug Reports:** Open issues for problems
- **Feature Requests:** Suggest new capabilities
- **Code Contributions:** Submit pull requests
- **Documentation:** Improve guides and examples
- **Testing:** Help with integration and end-to-end testing

## Development Setup

```
# Install development dependencies
uv sync --extra dev
```

```
# Run tests
```



# Support & Resources

## Documentation

- [README.md](#): Comprehensive setup guide
- [Architecture Guide](#): Technical architecture overview
- [User Guide](#): Complete user documentation
- [Configuration Guide](#): Configuration options and examples
- [Development Guide](#): Development setup and guidelines
- [Troubleshooting](#): Common issues and solutions
- [Data Processing Pipeline](#): Batch processing documentation
- **API Documentation**: Available at `/docs` when API server is running

## Community

- **GitHub**: <https://github.com/maximilian/ragme.js>

## Conclusion

RAGme.io represents the next generation of personal knowledge management:

-  **Intelligent Content Discovery:** Automatically process and index your content with AI analysis
-  **AI-Powered Insights:** Get intelligent responses from your personal knowledge base with caching
-  **Seamless Integration:** Multiple ways to add and interact with content across languages
-  **Scalable Architecture:** Built for growth with Kubernetes deployment and horizontal scaling
-  **Modern Interface:** Beautiful, responsive web interface with real-time features and OAuth
-  **Flexible Deployment:** Support for multiple vector databases and production

# Development & Configuration

## Development Environment

```
# Setup development environment
./setup.sh --force
source .venv/bin/activate
uv sync --extra dev

# Development workflow
./start.sh                # Start all services
./start.sh compile-frontend # Compile frontend after changes
./start.sh restart-backend  # Restart API, MCP, Agent
./start.sh restart-frontend  # Restart frontend only

# Monitor logs during development
./tools/tail-logs.sh all
```

## Comprehensive Testing

*Thank you for your attention! Questions and feedback welcome.*